

Department of Artificial Intelligence & Data Science
School of Computing, FAST-NUCES

Degree: DS/AI

Total Marks: 20

Instructor: Dr. Zohair Ahmed

Due Date: As Per GCR

Assignment 1: Understanding the Paper

Objective

- Develop a deep conceptual understanding of the **assigned project** and its **associated research paper**.
- The project and **three related** research papers have already been assigned to each group. Please check the group list to see the project allocated to your group. From the three papers listed for your project, each group **must select one paper** to study and analyze.
- If any of the provided **papers does not include a proposed model or dataset**, you must inform the teacher before finalizing your selection. If you select such a paper without informing the teacher, **any resulting issues will be the responsibility of the group**.

Important Policy

If a student submits work that appears to be **copied from AI tools or external sources and cannot explain it**, the instructor may assign **zero marks for the project**.

Therefore, please ensure that:

- you actively participate in the project
- you understand the work being submitted
- you maintain academic integrity

Tasks

Students must perform a **systematic paper analysis**:

1. Paper Context

- Research problem addressed
- Motivation and real-world importance
- Narrow Target task

2. Literature Context

- All Prior methods referenced

- How the proposed approach differs from earlier work

3. Methodology Breakdown

Explain clearly:

- Model architecture
- Dataset(s) used
- Training procedure
- Evaluation metrics
- Baselines used for comparison

Students should include **methodology diagrams if available**.

4. Results Analysis

Discuss:

- Main experimental results
- Performance comparison with baselines
- Ablation studies
- Error analysis (if present)

5. Critical Analysis

Students must identify:

- Limitations of the method
- Dataset bias or limitations
- Reproducibility concerns
- Scalability issues

6. Research Opportunities

Propose **2–3 possible extensions**, such as:

- new dataset
- multilingual adaptation
- model modification
- efficiency improvements

Deliverables

Report Length: 5–7 pages

Structure:

- 1. Introduction
- 2. Paper Summary
- 3. Methodology Analysis
- 4. Experimental Setup
- 5. Critical Analysis
- 6. Future Research Opportunities
- 7. References

Evaluation Criteria (Total 100)

Criteria	Weight
Depth of understanding	30%
Quality of analysis	25%
Critical thinking	20%
Clarity of writing	15%
Citation & academic integrity	10%